

RESEARCHES ON THE PRIME DIVISORS OF A CLASS OF FORMULAE OF THE FOURTH DEGREE.

First memoir.

In the literary announcements of Göttingen (11 April 1825) one finds the extract of a memoir of indeterminate analysis which Mr. Gauss presented to the Royal Society of Sciences of that city, but which has not yet been printed. This memoir is the first of a series of memoirs which the illustrious author of the *Disquisitiones arithmeticae* proposes to give on the theory of biquadratic residues, and its object is to determine the distinctive characters of the prime divisors of the formula $x^4 - 2$. The author establishes in it two extremely elegant theorems which can serve to decide whether a prime number, divisor of $x^2 - 2$, divides or does not divide the preceding formula. Having learned, in the course of the year just ended, of the cited extract, which contains only the statements of the two theorems just mentioned and of some auxiliary propositions, I wished to demonstrate on my side the beautiful theorems discovered by Mr. Gauss. The researches which I made with this aim led me to find a proof founded on extremely simple considerations and probably entirely different from that of Mr. Gauss, which appears to require very delicate and rather extensive preliminary researches. I then applied analogous considerations to other questions and particularly to the search for the properties which distinguish the prime divisors of the formula $\alpha x^4 + \beta x^2 + \gamma$; and I thus arrived at a large number of interesting theorems. The rapid exposition of part of the results to which these researches have led me is the object of the present memoir. I begin by setting down some definitions and by stating some theorems very easy to establish and on which we shall have to rely in what follows.

1.

"If one can assign to the indeterminate x a value such that $x^4 - A$ becomes divisible by B , A will be called a biquadratic residue with respect to B ."

It is easy to see that, if a number A is a biquadratic residue with respect to a number B , or in other words, if B is a divisor of $x^4 - A$, each prime factor of B will likewise be a divisor of $x^4 - A$, and reciprocally, that if this condition holds with respect to every prime factor of the number B , B will itself be a divisor of $x^4 - A$. Thus, when it is a question of assigning all the numbers which divide the formula $x^4 - A$, one can confine oneself to considering prime numbers.

It is no less evident that, for a number to divide the formula $x^4 - A$, it is necessary that this number be a divisor of $x^2 - A$. There are therefore only the prime divisors of this latter formula to examine, that is to say, the prime numbers with respect to which A is a quadratic residue. One may add that, relative to those among these latter which are of the form $4n + 3$, the question presents no difficulty; for one verifies by a very simple reasoning that every prime number $4n + 3$, divisor of $x^2 - A$, also divides the formula $x^4 - A$.

Let p be a prime number $4n + 1$, divisor of $x^2 - A$, A designating a positive

or negative number not divisible by p ; one has, as is known,

$$A^{\frac{p-1}{2}} \equiv 1 \pmod{p}.$$

From this one concludes

$$A^{\frac{p-1}{4}} \equiv \pm 1 \pmod{p},$$

and one proves easily that the upper or lower sign will hold according as p divides or does not divide the formula $x^4 - A$. Thus one may state this theorem:

" A designating a number which is a quadratic residue with respect to the prime number $p = 4n + 1$, one will have

$$A^{\frac{p-1}{4}} \equiv 1 \quad \text{or} \quad A^{\frac{p-1}{4}} \equiv -1 \pmod{p}.$$

In the first case, A will be a biquadratic residue with respect to p ; in the second, A will be a biquadratic non-residue with respect to p ."

If this theorem is applied to the case where $A = -1$, one will find that -1 is a biquadratic residue with respect to the prime numbers $8n + 1$, and on the contrary a non-residue relative to those of the form $8n + 5$. From the preceding theorem one easily deduces the following other theorem, in whose statement one supposes, as in all that follows, that p is a prime number $4n + 1$, and that A and A' designate numbers not divisible by p and which are both quadratic residues with respect to p .

"If the numbers A and A' are both biquadratic residues or both biquadratic non-residues with respect to p , the product AA' will be a biquadratic residue with respect to p ; if, on the contrary, one of the numbers A and A' is a residue and the other a biquadratic non-residue relative to p , AA' will be a biquadratic non-residue with respect to p ."

Taking $A' = -1$, and having regard to what precedes, one will see that, if p is of the form $8n + 1$, A will at the same time as $-A$ be a biquadratic residue or non-residue, and that, on the contrary, if p is of the form $8n + 5$, one of the numbers A and $-A$ will be a residue and the other a biquadratic non-residue with respect to p .

2.

After establishing these preliminaries, we shall deal with the search for the characters which distinguish the prime divisors of the formula $x^4 - 2$. It is known that the prime divisors of $x^2 - 2$ are of one of these two forms: $8n + 1$, $8n + 7$, and that reciprocally every prime number of one of these forms divides the formula $x^2 - 2$. According to what we said in the preceding paragraph, we need not take account of the latter of these two forms, and it will suffice to consider the prime numbers $8n + 1$. Let p be a prime number of this kind, and put, as one is permitted to do,

$$p = t^2 + 2u^2,$$

where u will be even and t odd. Let $u = 2^{\nu} k k' k'' \dots$, 2^{ν} being the highest power of 2 which divides u , and $k, k', k'', \dots$ designating odd prime numbers, several of which may be equal to one another. The equation $t^2 + 2u^2 = p$ immediately gives

$t^2 \equiv p \pmod{k}$. The number p is therefore a quadratic residue with respect to k , which we shall write thus:

$$\left(\frac{p}{k}\right) = 1,$$

adopting the notation used by Mr. Legendre. Recall that, if c designates a prime number and M any number not divisible by c , this illustrious geometer uses the sign $\left(\frac{M}{c}\right)$ to designate the remainder which one obtains by dividing by c the power $M^{\frac{c-1}{2}}$, a remainder which is known to be equal to 1 or -1 , according as M is or is not a quadratic residue with respect to c . Applying to the relation $\left(\frac{p}{k}\right) = 1$ the theorem known under the name law of reciprocity (the *theorema fondamentale* of Mr. Gauss), one will have, p being of the form $4n + 1$,

$$\left(\frac{k}{p}\right) = 1.$$

One finds in the same way:

$$\left(\frac{k'}{p}\right) = 1, \quad \left(\frac{k''}{p}\right) = 1, \quad \text{etc.}$$

On the other hand, since p is of the form $8n + 1$, one also has $\left(\frac{2}{p}\right) = 1$, and consequently $\left(\frac{2^\nu}{p}\right) = 1$. Multiplying this last relation by all the preceding ones, one obtains:

$$\left(\frac{2^\nu k k' k'' \dots}{p}\right) = \left(\frac{u}{p}\right) = 1.$$

Now consider the simple factors of the odd number t , which we shall divide into two classes. The first class will comprise the prime divisors of one of these two forms, $8n + 1$, $8n + 7$, and the numbers belonging to it will be designated by $g, g', g'', \dots$; the second class will be composed of numbers $h, h', h'', \dots$ contained in these two forms, $8n + 3$, $8n + 5$. One first has:

$$t = gg'g'' \dots \times hh'h'' \dots,$$

and one will then conclude from the equation $p = t^2 + 2u^2$:

$$\left(\frac{2p}{g}\right) = 1, \quad \left(\frac{2p}{g'}\right) = 1, \quad \left(\frac{2p}{g''}\right) = 1, \quad \text{etc.}$$

$$\left(\frac{2p}{h}\right) = 1, \quad \left(\frac{2p}{h'}\right) = 1, \quad \left(\frac{2p}{h''}\right) = 1, \quad \text{etc.}$$

On the other hand, by known theorems one has:

$$\left(\frac{2}{g}\right) = 1, \quad \left(\frac{2}{g'}\right) = 1, \quad \left(\frac{2}{g''}\right) = 1, \quad \text{etc.}$$

$$\left(\frac{2}{h}\right) = -1, \quad \left(\frac{2}{h'}\right) = -1, \quad \left(\frac{2}{h''}\right) = -1, \quad \text{etc.}$$

Comparing these relations with the preceding ones, one finds:

$$\left(\frac{p}{g}\right) = 1, \quad \left(\frac{p}{g'}\right) = 1, \quad \left(\frac{p}{g''}\right) = 1, \quad \text{etc.}$$

$$\left(\frac{p}{h}\right) = -1, \quad \left(\frac{p}{h'}\right) = -1, \quad \left(\frac{p}{h''}\right) = -1, \quad \text{etc.}$$

The application of the law of reciprocity to these last relations gives:

$$\left(\frac{g}{p}\right) = 1, \quad \left(\frac{g'}{p}\right) = 1, \quad \left(\frac{g''}{p}\right) = 1, \quad \text{etc.}$$

$$\left(\frac{h}{p}\right) = -1, \quad \left(\frac{h'}{p}\right) = -1, \quad \left(\frac{h''}{p}\right) = -1, \quad \text{etc.}$$

Hence, on multiplying,

$$\left(\frac{gg'g'' \cdots \times hh'h'' \cdots}{p}\right) = \left(\frac{t}{p}\right) = \pm 1,$$

where the upper or lower sign must be taken according as the numbers $h, h', h'', \dots$ are even or odd in number. But it is easy to see, from the equation $t = gg'g'' \cdots \times hh'h'' \cdots$, that the first case will occur when t is of one of these forms, $8n+1, 8n+7$, and the second when t is contained in one of these, $8n+3, 8n+5$. Therefore

$$\begin{aligned} \left(\frac{t}{p}\right) &= 1, & \text{when } t &= 8n+1 \text{ or } 8n+7, \\ \left(\frac{t}{p}\right) &= -1, & \text{when } t &= 8n+3 \text{ or } 8n+5. \end{aligned}$$

Let us return to the equation $t^2 + 2u^2 = p$ and put it in the form of a congruence:

$$t^2 \equiv -2u^2 \pmod{p}.$$

Raising both sides to the power $\frac{p-1}{4}$, one will have ($\frac{p-1}{4}$ being even):

$$t^{\frac{p-1}{2}} \equiv 2^{\frac{p-1}{4}} u^{\frac{p-1}{2}} \pmod{p},$$

or, what comes to the same thing, having proved that $\left(\frac{u}{p}\right) = 1$:

$$t^{\frac{p-1}{2}} \equiv 2^{\frac{p-1}{4}} \pmod{p}.$$

Thus one sees that ± 2 (the double sign may be put, since $p = 8n+1$) is or is not a biquadratic residue with respect to p according as one has $\left(\frac{t}{p}\right) = 1$ or $\left(\frac{t}{p}\right) = -1$. Comparing this result with what precedes, one has the theorem:

“ p designating a prime number $8n+1$, if one writes $p = t^2 + 2u^2$, I say that ± 2 will or will not be a biquadratic residue with respect to p according as t is of one of these forms, $8n+1, 8n+7$, or of one of these, $8n+3, 8n+5$.”

This is the first of the two theorems of Mr. Gauss mentioned in the preamble of this memoir. The second of these theorems is relative to the decomposition of the number p into two squares, and can easily be deduced from the one just established.

Let $p = \varphi^2 + \psi^2$ (where ψ is supposed divisible by 4) and equate this value of p with the one we have just considered. We shall thus have

$$p = t^2 + 2u^2 = \varphi^2 + \psi^2,$$

and, after transposing,

$$t^2 - \psi^2 = (t + \psi)(t - \psi) = \varphi^2 - 2u^2.$$

Since φ is odd, the greatest common divisor of φ and u will be odd; designate it by m and put $\varphi = m\varphi'$, $u = mu'$. Substitution of these values in the preceding equation changes it into

$$(t + \psi)(t - \psi) = m^2(\varphi'^2 - 2u'^2).$$

The odd number $t + \psi$ is thus composed of two factors E and K , the first of which divides m^2 , the second $\varphi'^2 - 2u'^2$; and the same is true of $t - \psi$, whose factors we denote by F and L , L being allowed to be negative. We therefore have

$$\begin{aligned} t + \psi &= EK, & t - \psi &= FL, \\ m^2 &= EF, & \varphi'^2 - 2u'^2 &= KL. \end{aligned}$$

It is easy to verify that the numbers E and F are relatively prime. Indeed, let δ be a prime divisor of E , and suppose that δ also divides F . The number δ would be a common divisor of $t + \psi$ and $t - \psi$, and consequently would divide the number t , which is the half-sum of the preceding two numbers. On the other hand, from the fact that δ is a prime divisor of E , it follows successively, having regard to the equations $EF = m^2$, $u = mu'$, that m^2 , m , and u are multiples of δ . The numbers t and u would therefore have the common divisor δ , and $p = t^2 + 2u^2$ would not be a prime number.

The product of the numbers E and F , which are relatively prime, being a square, each of them is also a square. Let $E = e^2$; then $t + \psi = e^2K$. The odd square e^2 is of the form $8n + 1$, and K , as an odd divisor of $\varphi'^2 + 2u'^2$ (where φ' and u' are relatively prime), is of one of these forms, $8n + 1$, $8n + 7$. Thus the number $t + \psi$ will itself be of one of the forms $8n + 1$, $8n + 7$. The number ψ , which we know to be divisible by 4, will be of the form $8n$ or of the form $8n + 4$. It follows from the preceding that, in the first case, t will be of one of these two forms, $8n + 1$, $8n + 7$, and in the second of one of these, $8n + 3$, $8n + 5$. Comparing this result with the preceding theorem, one has this other theorem:

“ p designating a prime number $8n + 1$, and having written $p = \varphi^2 + \psi^2$ (where ψ is supposed divisible by 4), ± 2 will or will not be a biquadratic residue with respect to p according as ψ is of the form $8n$ or of the form $8n + 4$.”

There is a third theorem suitable for deciding whether ± 2 is or is not a biquadratic residue relative to a prime number $p = 8n + 1$, which may be stated as follows:

“Having made, in any manner, $p = t^2 - 2u^2$, ± 2 will or will not be a biquadratic residue with respect to p according as t is of one of the forms $8n + 1$, $8n + 3$, or of one of these, $8n + 5$, $8n + 7$.”

We shall not stop to prove this theorem, which can be established directly and by considerations analogous to those on which the proof of the first of the two preceding theorems is founded. One can also deduce it from each of the preceding theorems almost as one has just passed from the first to the second.

We now pass to more general considerations. Let b be a prime number $4n + 3$, p a prime number $4n + 1$ capable of being put under the form $t^2 - bu^2$, and let us propose to decide whether $-b$ is or is not a biquadratic residue with respect to p . It is easy to see that, if p can be put under the form $t^2 - bu^2$, one can always do so in such a way that u is even, and consequently t odd¹. Let therefore $p = t^2 - bu^2$, and put $u = 2^\nu k k' k'' \dots$, where $k, k', k'', \dots$ are the simple odd factors of u . One concludes immediately from the preceding equation:

$$\left(\frac{p}{k}\right) = 1, \quad \left(\frac{p}{k'}\right) = 1, \quad \left(\frac{p}{k''}\right) = 1, \quad \text{etc.}$$

The application of the reciprocity law then gives, p being of the form $4n + 1$:

$$\left(\frac{k}{p}\right) = 1, \quad \left(\frac{k'}{p}\right) = 1, \quad \left(\frac{k''}{p}\right) = 1, \quad \text{etc.}$$

Multiplying these relations together and with the identical relation $\left(\frac{2^\nu}{p}\right) = \left(\frac{2^\nu}{p}\right)$, one has the result

$$(\alpha) \quad \left(\frac{u}{p}\right) = \left(\frac{2^\nu}{p}\right).$$

Now decompose the odd number t into its simple factors and divide these factors into two classes. Those of the first class will be designated by $g, g', g'', \dots$, and are such that

$$(\beta) \quad \left(\frac{-b}{g}\right) = 1, \quad \left(\frac{-b}{g'}\right) = 1, \quad \left(\frac{-b}{g''}\right) = 1, \quad \text{etc.}$$

As for those which form the second class and which we shall designate by $h, h', h'', \dots$, they are such that

$$(\beta') \quad \left(\frac{-b}{h}\right) = -1, \quad \left(\frac{-b}{h'}\right) = -1, \quad \left(\frac{-b}{h''}\right) = -1, \quad \text{etc.}$$

The product of all these numbers is equal to t , that is,

$$t = gg'g'' \dots \times hh'h'' \dots.$$

Inspection of the equation $t^2 - bu^2 = p$ gives the results

$$\left(\frac{-bp}{g}\right) = 1, \quad \left(\frac{-bp}{g'}\right) = 1, \quad \left(\frac{-bp}{g''}\right) = 1, \quad \text{etc.}$$

¹To prove that this is always possible, we shall show that, if one has $t^2 - bu^2 = p$, t being even, it is easy to deduce from the values of t and u other numbers t' (odd) and u' which likewise satisfy the equation $t'^2 - bu'^2 = p$. Let r and s be the least numbers such that $r^2 - bs^2 = 1$; I say that r will be even. Indeed, it is known that, if b designates a prime number $4n + 3$, the equation $\rho^2 - b\sigma^2 = \pm 2$ is always possible, and that, ρ and σ being supposed the least numbers which satisfy it, one has $r = \rho^2 \mp 1$, $s = \rho\sigma$ (Theory of Numbers, nos. 44, 45). From this, and from the fact that the numbers ρ and σ are evidently both odd, it follows that r is an even number. This being so, if one multiplies together the equations $r^2 - bs^2 = 1$, $t^2 - bu^2 = p$, one obtains $(rt + bsu)^2 - b(ru + st)^2 = p$, and one will easily see that $rt \pm bsu$ is an odd number.

$$\left(\frac{-bp}{h}\right) = 1, \quad \left(\frac{-bp}{h'}\right) = 1, \quad \left(\frac{-bp}{h''}\right) = 1, \quad \text{etc.}$$

Comparison of these relations with the preceding ones then gives

$$\left(\frac{p}{g}\right) = 1, \quad \left(\frac{p}{g'}\right) = 1, \quad \left(\frac{p}{g''}\right) = 1, \quad \text{etc.}$$

$$\left(\frac{p}{h}\right) = -1, \quad \left(\frac{p}{h'}\right) = -1, \quad \left(\frac{p}{h''}\right) = -1, \quad \text{etc.}$$

Applying now the reciprocity law, one obtains

$$\left(\frac{g}{p}\right) = 1, \quad \left(\frac{g'}{p}\right) = 1, \quad \left(\frac{g''}{p}\right) = 1, \quad \text{etc.}$$

$$\left(\frac{h}{p}\right) = -1, \quad \left(\frac{h'}{p}\right) = -1, \quad \left(\frac{h''}{p}\right) = -1, \quad \text{etc.}$$

Multiplying these relations together, one will have

$$\left(\frac{gg'g'' \cdots \times hh'h'' \cdots}{p}\right) = \left(\frac{t}{p}\right) = \pm 1,$$

the upper or lower sign holding according as the numbers $h, h', h'', \dots$ are even or odd in number. On the other hand, if one applies the reciprocity law to the relations (β) and (β') , one obtains, b being of the form $4n + 3$,

$$\left(\frac{g}{b}\right) = 1, \quad \left(\frac{g'}{b}\right) = 1, \quad \left(\frac{g''}{b}\right) = 1, \quad \text{etc.}$$

$$\left(\frac{h}{b}\right) = -1, \quad \left(\frac{h'}{b}\right) = -1, \quad \left(\frac{h''}{b}\right) = -1, \quad \text{etc.}$$

equalities which, when multiplied together, give

$$\left(\frac{gg'g'' \cdots \times hh'h'' \cdots}{b}\right) = \left(\frac{t}{b}\right) = \pm 1,$$

where the upper or lower sign must be taken according as the numbers $h, h', h'', \dots$ are even or odd in number.

Comparison of this result with the one obtained a moment ago shows that one always has

$$(\gamma) \quad \left(\frac{t}{p}\right) = \left(\frac{t}{b}\right).$$

Let us now return to the equation $p = t^2 - bu^2$ and put it in congruence form:

$$t^2 \equiv bu^2 \pmod{p}.$$

From this, raising both sides to the power $\frac{p-1}{4}$, one obtains

$$t^{\frac{p-1}{2}} \equiv b^{\frac{p-1}{4}} u^{\frac{p-1}{2}} \pmod{p}.$$

The formula (α) , namely $\left(\frac{u}{p}\right) = \left(\frac{2^\nu}{p}\right)$, where 2^ν denotes the highest power dividing u , can be presented in another manner. For this it is necessary to distinguish two cases, according as p is of the form $8n + 1$ or of the form $8n + 5$.

If p is a prime number $8n+1$, one has, as is known, $\left(\frac{2}{p}\right) = 1$, and consequently $\left(\frac{2^\nu}{p}\right) = 1$; the formula in question therefore becomes in this case $\left(\frac{u}{p}\right) = 1$. If p is a prime number $8n+5$, the number ν is equal to 1; for if ν were greater than unity, u^2 would be divisible by 8, and $p = t^2 - bu^2$ would be of the form $8n+1$. Moreover, since in this case $\left(\frac{2}{p}\right) = -1$, formula (α) becomes $\left(\frac{u}{p}\right) = -1$. The two cases just examined are included in the formula

$$\left(\frac{u}{p}\right) = (-1)^{\frac{p-1}{4}},$$

which is equivalent to the congruence

$$u^{\frac{p-1}{2}} \equiv (-1)^{\frac{p-1}{4}} \pmod{p}.$$

Substituting the value which this gives for $u^{\frac{p-1}{2}}$ in the congruence obtained above, one has

$$t^{\frac{p-1}{2}} \equiv (-b)^{\frac{p-1}{4}} \pmod{p},$$

a result which shows that $-b$ will or will not be a biquadratic residue with respect to p according as t is or is not a quadratic residue with respect to p . Comparing this result with that contained in formula (γ) , one arrives at the theorem which we shall state:

“Let b designate a prime number $4n+3$, and p a prime number $4n+1$ capable of being put under the form $t^2 - bu^2$. Having written $p = t^2 - bu^2$ (where t is supposed odd), I say that $-b$ will or will not be a biquadratic residue with respect to p according as t is or is not a quadratic residue with respect to b .”

4.

We shall now deduce from this theorem another by means of which one can decide still more quickly whether $-b$ is or is not a biquadratic residue with respect to p . Preserve the preceding notations and put $p = \varphi^2 + \psi^2$ (where ψ is supposed even). Equating this value of p to the one considered above, one has

$$p = t^2 - bu^2 = \varphi^2 + \psi^2,$$

and, after transposing,

$$t^2 - \psi^2 = (t + \psi)(t - \psi) = \varphi^2 + bu^2.$$

There are now two cases to distinguish, according as φ is or is not divisible by b . We begin with the latter of these two cases. Let m be the greatest common divisor of φ and u , which will be odd (φ being odd) and not divisible by b , and put $\varphi = m\varphi'$, $u = mu'$. Substitution of these values in the last equation changes it into

$$(t + \psi)(t - \psi) = m^2(\varphi'^2 + bu'^2).$$

It is evident, from this equation, that $t + \psi$ is composed of two factors E and K , the first of which divides m^2 , the second $\varphi'^2 + bu'^2$, and that the same is true of $t - \psi$. Denoting the factors of this last number by F and L , we shall have the equations

$$\begin{aligned} t + \psi &= EK, & t - \psi &= FL, \\ m^2 &= EF, & \varphi'^2 + bu'^2 &= KL. \end{aligned}$$